

superio

An autonomous options agent where
the model never touches the limits.

Defined-risk option structures on SPY, QQQ and IWM • Alpaca paper trading

THE PROBLEM

Everyone lets the model size the bet. That's the part it is worst at.

In Nof1's Alpha Arena, four of six language models lost money trading live. GPT-5 finished down 63%, Gemini down 57%, Grok down 58%. The winner's edge was sizing discipline, not prediction.

Independent replication of LLM trading research finds the edge disappears out of sample. A 2026 audit rated zero of nineteen such studies at the top reproducibility tier.

MODELS ARE GOOD AT

reading

Regime narration, news, choosing between framed options.

MODELS ARE BAD AT

arithmetic

Position sizing, strike selection, respecting a limit under pressure.

The model returns an index.

Deterministic code builds every candidate structure from the live chain, sizes it, and refuses anything failing a gate. Only then is the model asked to choose.

```
{"action": "trade", "refs": ["QQQ:call_debit_spread"], "confidence": 0.8}
```

That is its entire output space. It cannot invent a structure, move a strike, change a quantity, or widen a limit, because none of those are things it can say.

The worst a confused or prompt-injected model can do is pick a slightly worse safe trade, or refuse to trade.

Four roles. One of them is allowed to be wrong.



Seven gates. Numbered, tested, journalled by name.

	GATE	LIMIT
G1	Kill switches	-3% on the day flattens · -8% drawdown ends the event
G2	Trade budget	8 new structures a day, 10 open at once
G3	Defined risk	Every short leg needs same-expiry cover. No flag disables it
G4	Leg liquidity	Bid/ask within 12% of mid, per leg
G5	Pricing	Sell no spread under 18% of width · pay no more than 45%
G6	Event blackout	No writing premium across a scheduled catalyst
G7	Sizing	0.75% of equity per structure · 6% total · 3% convex

Priced at the touch, not the mid. Alpaca's paper engine fills against the NBB0, so sizing off mid prices understates real maximum loss.

We show you the trades it refused.

```
risk_officer  SPY  rejected  G5 pricing: credit 17.0% of width  
                                     is below the 18% floor  
risk_officer  IWM  rejected  G5 pricing: credit 13.0% of width  
risk_officer  QQQ  approved  G3 all short legs covered by  
                                     same-expiry long legs
```

Every refusal is attributed to the gate that caused it and published on the live dashboard beside the trades that passed.

An agent that shows only its trades is showing **half its behaviour** – and the half it refused is the half that explains the risk.

Both interfaces, split on purpose.

CLI — EXECUTION

reproducible

Every order is `alpaca api POST /v2/orders` with a JSON body, stored in the journal beside the proposal and the risk verdict that produced it. A judge can paste any trade into a terminal.

MCP — RESEARCH

read-only

41 tools over one stdio session for news and market state, with output wrapped in a trust envelope marking it as data rather than instructions.

Execution deliberately avoids MCP: its v2 multi-leg order bug (issue #97) sends the legs array as a raw string, and every structure this agent trades is multi-leg.

Verified end to end: a spread opened at a net 1.05 debit and closed as a package for a net 1.05 credit.

RESULTS

Three variants, one tape.

Four and a half sessions cannot separate skill from variance. So rather than run one account and tell a story about the number, the same engine runs three presets side by side and the week produces a comparison.

BARBELL – JUDGED

PA3Z3MMXLBZ0

Defined-risk premium plus a capped convex sleeve, routed by the volatility signal.

CONVEX TILT

PA3YJ4131VZX

Convexity-led, larger convex cap, lower short delta.

INCOME ONLY – CONTROL

PA3U027VIRFW

Premium selling only, no convex sleeve.

Live dashboard • realized P&L, win rate, profit factor, drawdown, and every refusal

What we would not claim.

Paper fills are more generous than live ones, especially on wide quotes, so any result here is optimistic by construction. The free data feed delays trades by fifteen minutes.

No historical option prices are available on this plan, so the replay harness prices entry and exit with the same model and simulates no volatility risk. It validates mechanics, not edge – and we say so in the tool's own output.

Four and a half sessions is far too short a sample to distinguish skill from luck. We would say that whichever way the number lands on Friday.

TESTS

34

gates, order ladder, every exit rule

NAKED SHORT EXPOSURE

structurally impossible

not a setting

LICENCE

MIT

public repo, commits across the event